Data understading and Cleaning Using Python

In [2]:
```
import pandas as pd
```

In [3]:
```
data = pd.read_csv("C:\\Users\\avant\\Downloads\\OneDrive\\Documents\\TelecomCustomerChurn\\telecom_customer_churn.csv")
```

In [7]:
```
#Understanding the data
```

In [13]:
```
data.shape
```

Out[13]:
```
(7043, 38)
```

In [8]:
```
data.head()
```

Out[8]:

	Customer ID	Gender	Age	Married	Number of Dependents	City	Zip Code	Latitude	Longitude	Number of Referrals	...	Payment Method	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	Customer Status	Churn Category	Churn Reason
0	0002-ORFBO	Female	37	Yes	0	Frazier Park	93225	34.827662	-118.999073	2	...	Credit Card	65.6	593.30	0.00	0	381.51	974.81	Stayed	NaN	NaN
1	0003-MKNFE	Male	46	No	0	Glendale	91206	34.162515	-118.203869	0	...	Credit Card	-4.0	542.40	38.33	10	96.21	610.28	Stayed	NaN	NaN
2	0004-TLHLJ	Male	50	No	0	Costa Mesa	92627	33.645672	-117.922613	0	...	Bank Withdrawal	73.9	280.85	0.00	0	134.60	415.45	Churned	Competitor	Competitor had better devices
3	0011-IGKFF	Male	78	Yes	0	Martinez	94553	38.014457	-122.115432	1	...	Bank Withdrawal	98.0	1237.85	0.00	0	361.66	1599.51	Churned	Dissatisfaction	Product dissatisfaction
4	0013-EXCHZ	Female	75	Yes	0	Camarillo	93010	34.227846	-119.079903	3	...	Credit Card	83.9	267.40	0.00	0	22.14	289.54	Churned	Dissatisfaction	Network reliability

5 rows × 38 columns

In [9]:
```
data.tail()
```

Out[9]:

	Customer ID	Gender	Age	Married	Number of Dependents	City	Zip Code	Latitude	Longitude	Number of Referrals	...	Payment Method	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	Customer Status	Churn Category	Churn Reason
7038	9987-LUTYD	Female	20	No	0	La Mesa	91941	32.759327	-116.997260	0	...	Credit Card	55.15	742.90	0.0	0	606.84	1349.74	Stayed	NaN	NaN
7039	9992-RRAMN	Male	40	Yes	0	Riverbank	95367	37.734971	-120.954271	1	...	Bank Withdrawal	85.10	1873.70	0.0	0	356.40	2230.10	Churned	Dissatisfaction	Product dissatisfaction
7040	9992-UJOEL	Male	22	No	0	Elk	95432	39.108252	-123.645121	0	...	Credit Card	50.30	92.75	0.0	0	37.24	129.99	Joined	NaN	NaN
7041	9993-LHIEB	Male	21	Yes	0	Solana Beach	92075	33.001813	-117.263628	5	...	Credit Card	67.85	4627.65	0.0	0	142.04	4769.69	Stayed	NaN	NaN
7042	9995-HOTOH	Male	36	Yes	0	Sierra City	96125	39.600599	-120.636358	1	...	Bank Withdrawal	59.00	3707.60	0.0	0	0.00	3707.60	Stayed	NaN	NaN

5 rows × 38 columns

In [10]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 38 columns):
 #   Column                             Non-Null Count  Dtype  
---  ------                             --------------  -----  
 0   Customer ID                        7043 non-null   object 
 1   Gender                             7043 non-null   object 
 2   Age                                7043 non-null   int64  
 3   Married                            7043 non-null   object 
 4   Number of Dependents               7043 non-null   int64  
 5   City                               7043 non-null   object 
 6   Zip Code                           7043 non-null   int64  
 7   Latitude                           7043 non-null   float64
 8   Longitude                          7043 non-null   float64
 9   Number of Referrals                7043 non-null   int64  
 10  Tenure in Months                   7043 non-null   int64  
 11  Offer                              7043 non-null   object 
 12  Phone Service                      7043 non-null   object 
 13  Avg Monthly Long Distance Charges  6361 non-null   float64
 14  Multiple Lines                     6361 non-null   object 
 15  Internet Service                   7043 non-null   object 
 16  Internet Type                      5517 non-null   object 
 17  Avg Monthly GB Download            5517 non-null   float64
 18  Online Security                    5517 non-null   object 
 19  Online Backup                      5517 non-null   object 
 20  Device Protection Plan             5517 non-null   object 
 21  Premium Tech Support               5517 non-null   object 
 22  Streaming TV                       5517 non-null   object 
 23  Streaming Movies                   5517 non-null   object 
 24  Streaming Music                    5517 non-null   object 
 25  Unlimited Data                     5517 non-null   object 
 26  Contract                           7043 non-null   object 
 27  Paperless Billing                  7043 non-null   object 
 28  Payment Method                     7043 non-null   object 
 29  Monthly Charge                     7043 non-null   float64
 30  Total Charges                      7043 non-null   float64
 31  Total Refunds                      7043 non-null   float64
 32  Total Extra Data Charges           7043 non-null   int64  
 33  Total Long Distance Charges        7043 non-null   float64
 34  Total Revenue                      7043 non-null   float64
 35  Customer Status                    7043 non-null   object 
 36  Churn Category                     1869 non-null   object 
 37  Churn Reason                       1869 non-null   object 
dtypes: float64(9), int64(6), object(23)
memory usage: 2.0+ MB
```

In [11]:
```
data.describe()
```

Out[11]:

	Age	Number of Dependents	Zip Code	Latitude	Longitude	Number of Referrals	Tenure in Months	Avg Monthly Long Distance Charges	Avg Monthly GB Download	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue
count	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	6361.000000	5517.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000	7043.000000
mean	46.509726	0.468692	93486.070567	36.197455	-119.756684	1.951867	32.386767	25.420517	26.189958	63.596131	2280.381264	1.962182	6.860713	749.099262	3034.379056
std	16.750352	0.962802	1856.767505	2.468929	2.154425	3.001199	24.542061	14.200374	19.586585	31.204743	2266.220462	7.902614	25.104978	846.660055	2865.204542
min	19.000000	0.000000	90001.000000	32.555828	-124.301372	0.000000	1.000000	1.010000	2.000000	-10.000000	18.800000	0.000000	0.000000	0.000000	21.360000
25%	32.000000	0.000000	92101.000000	33.990646	-121.788090	0.000000	9.000000	13.050000	13.000000	30.400000	400.150000	0.000000	0.000000	70.545000	605.610000
50%	46.000000	0.000000	93518.000000	36.205465	-119.595293	0.000000	29.000000	25.690000	21.000000	70.050000	1394.550000	0.000000	0.000000	401.440000	2108.640000
75%	60.000000	0.000000	95329.000000	38.161321	-117.969795	3.000000	55.000000	37.680000	30.000000	89.750000	3786.600000	0.000000	0.000000	1191.100000	4801.145000
max	80.000000	9.000000	96150.000000	41.962127	-114.192901	11.000000	72.000000	49.990000	85.000000	118.750000	8684.800000	49.790000	150.000000	3564.720000	11979.340000

In [12]:
```python
data.describe(include='object')
```

Out[12]:

	Customer ID	Gender	Married	City	Offer	Phone Service	Multiple Lines	Internet Service	Internet Type	Online Security	...	Streaming TV	Streaming Movies	Streaming Music	Unlimited Data	Contract	Paperless Billing	Payment Method	Customer Status	Churn Category	Churn Reason
count	7043	7043	7043	7043	7043	7043	6361	7043	5517	5517	...	5517	5517	5517	5517	7043	7043	7043	7043	1869	1869
unique	7043	2	2	1106	6	2	2	2	3	2	...	2	2	2	2	3	2	3	3	5	20
top	0002-ORFBO	Male	No	Los Angeles	None	Yes	No	Yes	Fiber Optic	No	...	No	No	No	Yes	Month-to-Month	Yes	Bank Withdrawal	Stayed	Competitor	Competitor had better devices
freq	1	3555	3641	293	3877	6361	3390	5517	3035	3498	...	2810	2785	3029	4745	3610	4171	3909	4720	841	313

4 rows × 23 columns

In [15]:
```python
data.columns
```

Out[15]:
```
Index(['Customer ID', 'Gender', 'Age', 'Married', 'Number of Dependents',
       'City', 'Zip Code', 'Latitude', 'Longitude', 'Number of Referrals',
       'Tenure in Months', 'Offer', 'Phone Service',
       'Avg Monthly Long Distance Charges', 'Multiple Lines',
       'Internet Service', 'Internet Type', 'Avg Monthly GB Download',
       'Online Security', 'Online Backup', 'Device Protection Plan',
       'Premium Tech Support', 'Streaming TV', 'Streaming Movies',
       'Streaming Music', 'Unlimited Data', 'Contract', 'Paperless Billing',
       'Payment Method', 'Monthly Charge', 'Total Charges', 'Total Refunds',
       'Total Extra Data Charges', 'Total Long Distance Charges',
       'Total Revenue', 'Customer Status', 'Churn Category', 'Churn Reason'],
      dtype='object')
```

In [22]:
```python
data.isnull().sum()
```

Out[22]:
```
Customer ID                          0
Gender                               0
Age                                  0
Married                              0
Number of Dependents                 0
City                                 0
Zip Code                             0
Latitude                             0
Longitude                            0
Number of Referrals                  0
Tenure in Months                     0
Offer                                0
Phone Service                        0
Avg Monthly Long Distance Charges  682
Multiple Lines                     682
Internet Service                     0
Internet Type                     1526
Avg Monthly GB Download           1526
Online Security                   1526
Online Backup                     1526
Device Protection Plan            1526
Premium Tech Support              1526
Streaming TV                      1526
Streaming Movies                  1526
Streaming Music                   1526
Unlimited Data                    1526
Contract                             0
Paperless Billing                    0
Payment Method                       0
Monthly Charge                       0
Total Charges                        0
Total Refunds                        0
Total Extra Data Charges             0
Total Long Distance Charges          0
Total Revenue                        0
Customer Status                      0
Churn Category                    5174
Churn Reason                      5174
dtype: int64
```

In [19]:
```
data['Contract'].value_counts()
```

Out[19]:
```
Month-to-Month    3610
Two Year          1883
One Year          1550
Name: Contract, dtype: int64
```

In [20]:
```
data['Gender'].value_counts()
```

Out[20]:
```
Male      3555
Female    3488
Name: Gender, dtype: int64
```

In [21]:
```
data['Churn Category'].value_counts()
```

Out[21]:
```
Competitor         841
Dissatisfaction    321
Attitude           314
Price              211
Other              182
Name: Churn Category, dtype: int64
```

In [1]:
```
#Data Cleaning
```

In [4]:
```
data['Customer ID'].duplicated().sum()
```
Out[4]:
```
0
```

In [5]:
```
#I handled missing values based on business logic—for example, churn-related fields were null for retained customers,
#so I labeled them as 'No Churn'.

data['Churn Category'].fillna("No Churn", inplace=True)
data['Churn Reason'].fillna("No Churn", inplace=True)
```

In [7]:
```
data.head()
```
Out[7]:

	Customer ID	Gender	Age	Married	Number of Dependents	City	Zip Code	Latitude	Longitude	Number of Referrals	...	Payment Method	Monthly Charge	Total Charges	Total Refunds	Total Extra Data Charges	Total Long Distance Charges	Total Revenue	Customer Status	Churn Category	Churn Reason
0	0002-ORFBO	Female	37	Yes	0	Frazier Park	93225	34.827662	-118.999073	2	...	Credit Card	65.6	593.30	0.00	0	381.51	974.81	Stayed	No Churn	No Churn
1	0003-MKNFE	Male	46	No	0	Glendale	91206	34.162515	-118.203869	0	...	Credit Card	-4.0	542.40	38.33	10	96.21	610.28	Stayed	No Churn	No Churn
2	0004-TLHLJ	Male	50	No	0	Costa Mesa	92627	33.645672	-117.922613	0	...	Bank Withdrawal	73.9	280.85	0.00	0	134.60	415.45	Churned	Competitor	Competitor had better devices
3	0011-IGKFF	Male	78	Yes	0	Martinez	94553	38.014457	-122.115432	1	...	Bank Withdrawal	98.0	1237.85	0.00	0	361.66	1599.51	Churned	Dissatisfaction	Product dissatisfaction
4	0013-EXCHZ	Female	75	Yes	0	Camarillo	93010	34.227846	-119.079903	3	...	Credit Card	83.9	267.40	0.00	0	22.14	289.54	Churned	Dissatisfaction	Network reliability

5 rows × 38 columns

In [9]:
```
data.columns = data.columns.str.lower()
```

In [11]:
```
data.head()
```
Out[11]:

	customer id	gender	age	married	number of dependents	city	zip code	latitude	longitude	number of referrals	...	payment method	monthly charge	total charges	total refunds	total extra data charges	total long distance charges	total revenue	customer status	churn category	churn reason
0	0002-ORFBO	Female	37	Yes	0	Frazier Park	93225	34.827662	-118.999073	2	...	Credit Card	65.6	593.30	0.00	0	381.51	974.81	Stayed	No Churn	No Churn
1	0003-MKNFE	Male	46	No	0	Glendale	91206	34.162515	-118.203869	0	...	Credit Card	-4.0	542.40	38.33	10	96.21	610.28	Stayed	No Churn	No Churn
2	0004-TLHLJ	Male	50	No	0	Costa Mesa	92627	33.645672	-117.922613	0	...	Bank Withdrawal	73.9	280.85	0.00	0	134.60	415.45	Churned	Competitor	Competitor had better devices
3	0011-IGKFF	Male	78	Yes	0	Martinez	94553	38.014457	-122.115432	1	...	Bank Withdrawal	98.0	1237.85	0.00	0	361.66	1599.51	Churned	Dissatisfaction	Product dissatisfaction
4	0013-EXCHZ	Female	75	Yes	0	Camarillo	93010	34.227846	-119.079903	3	...	Credit Card	83.9	267.40	0.00	0	22.14	289.54	Churned	Dissatisfaction	Network reliability

5 rows × 38 columns

In [13]:
```
data.columns = data.columns.str.replace(" ", "_")
```

In [14]:
```
data.head()
```

Out[14]:

	customer_id	gender	age	married	number_of_dependents	city	zip_code	latitude	longitude	number_of_referrals	...	payment_method	monthly_charge	total_charges	total_refunds	total_extra_data_charges	total_lon
0	0002-ORFBO	Female	37	Yes	0	Frazier Park	93225	34.827662	-118.999073	2	...	Credit Card	65.6	593.30	0.00	0	
1	0003-MKNFE	Male	46	No	0	Glendale	91206	34.162515	-118.203869	0	...	Credit Card	-4.0	542.40	38.33	10	
2	0004-TLHLJ	Male	50	No	0	Costa Mesa	92627	33.645672	-117.922613	0	...	Bank Withdrawal	73.9	280.85	0.00	0	
3	0011-IGKFF	Male	78	Yes	0	Martinez	94553	38.014457	-122.115432	1	...	Bank Withdrawal	98.0	1237.85	0.00	0	
4	0013-EXCHZ	Female	75	Yes	0	Camarillo	93010	34.227846	-119.079903	3	...	Credit Card	83.9	267.40	0.00	0	

5 rows × 38 columns

In [15]:
```
data.drop(['offer', 'paperless_billing',"married","number_of_dependents"], axis=1, inplace=True)
```

In [18]:
```
# after all understanding and cleaning steps
data.to_csv(r"C:\Users\avant\Downloads\OneDrive\Documents\Telecom_cleaned_data.csv", index=False)
```